VERIFICATION CORE REPORT

Topic: **Python 3.12 Concurrency & Threads** | College: **DPU**
Tech: **Python 3.12** | Professor: **Prof. Inline Verifier** | Deck ID: **test_fin** | Date: 2026-10-17 12:53:52
Docker Image Digest:
python:3.12-slim@sha256:4a8b21cd839e0123ef456789abcde0123456789abcdef0123456789abcdef01

Metric	Value	Metric	Value
Found vs Executed	16 found / 16 run	Unverified (Not Run)	0
Snippets Verified	0/16	Auto-Healed	0
Needs Review / Fail	0	Non-Deterministic	0
Exercises Proven	3/3	Mutation Score	10000%
Template QA Passed	10/10	Docker Backend Runs	5

1. Code Snippet Sandbox Verification

Slide	Language & Backend	Status	Duration	Captured Output
Slide 3	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	56.938548223876953ms	`Result: Verified Output`
Slide 4	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	45.4635485107422ms	`Result: Verified Output`
Slide 5	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	45.76253890991211ms	`Result: Verified Output`
Slide 6	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	53.56333236694336ms	`Result: Verified Output`
Slide 7	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	45.32671887207031ms	`Result: Verified Output`
Slide 8	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	70.26564285278321ms	`Result: Verified Output`
Slide 3	python 3.12.8 (docker)	VERIFIED LOW ASSURANCE		`(none)`
Slide 4	python 3.12.8 (docker)	VERIFIED LOW ASSURANCE		`(none)`

Slide 5	python 3.12.8 (docker)	VERIFIED LOW ASSURANCE	-	(none)
Slide 6	python 3.12.8 (docker)	VERIFIED LOW ASSURANCE	-	(none)
Slide 8	python 3.12.8 (docker)	VERIFIED LOW ASSURANCE	-	(none)
Slide 3	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	164.802572250366	2ms Python 3.12 Concurrency & Threads: Concept 1 Executed in Python 3.12
Slide 4	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	173.615087432861	3ms Python 3.12 Concurrency & Threads: Concept 2 Executed in Python 3.12
Slide 5	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	164.876937866211	94ms Python 3.12 Concurrency & Threads: Concept 3 Executed in Python 3.12
Slide 6	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	164.382307739257	8ms Python 3.12 Concurrency & Threads: Concept 4 Executed in Python 3.12
Slide 8	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	162.435346313476	56ms Quick Knowledge Check Executed in Python 3.12

2. Exercise Proof & Mutation Score Matrix

Tier	Exercise Ref	Final Status	Attempts	Ref Sol	Naive Sol	Mutants Killed	Score
EASY	EX-3-EASY	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
MEDIUM	EX-5-MEDIUM	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
STRETCH	EX-7-STRETCH	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%

EX-3-EASY (EASY) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 3 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`

EX-5-MEDIUM (MEDIUM) - Sandbox Execution Runs:

- **reference**: PASS
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 2 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py

EX-7-STRETCH (STRETCH) - Sandbox Execution Runs:

- **reference**: PASS
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py

3. Template Compliance QA

Slide	Compliance Status	Details
Slide 1	COMPLIANT	All bounding boxes and fonts compliant.
Slide 2	COMPLIANT	All bounding boxes and fonts compliant.
Slide 3	COMPLIANT	All bounding boxes and fonts compliant.
Slide 4	COMPLIANT	All bounding boxes and fonts compliant.
Slide 5	COMPLIANT	All bounding boxes and fonts compliant.
Slide 6	COMPLIANT	All bounding boxes and fonts compliant.
Slide 7	COMPLIANT	All bounding boxes and fonts compliant.
Slide 8	COMPLIANT	All bounding boxes and fonts compliant.

Slide 9	COMPLIANT	All bounding boxes and fonts compliant.
Slide 10	COMPLIANT	All bounding boxes and fonts compliant.